

Student Performance Prediction System

IS 108 – Intelligence System | SY 2025–2026

Subject	IS 108 – Intelligence System
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Dataset	Kaggle – Student Performance (25,000 records)
Live App	https://prediction-model-is108-empang-daguio-pilapil.streamlit.app/
Tools Used	Python, Streamlit, scikit-learn, Pandas, Plotly
ML Algorithms	K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Artificial Neural Network (ANN)

1. What Is This System?

This is a web app that predicts how a student is likely to perform — whether they will Pass, be Average, or Fail — based on information like their study habits, attendance, and test scores.

It was built for IS 108 (Intelligence System) as a project that shows how machine learning can help teachers spot students who might need extra support before it's too late.

What it can do

- Load a real student dataset from Kaggle (25,000 records)
- Clean and prepare the data automatically
- Train three different prediction models
- Compare which model performs best
- Predict a student's performance based on inputs you type in
- Give recommendations depending on the predicted result

Live App

You can try the system directly in your browser — no installation needed:

<https://prediction-model-is108-empang-daguio-pilapil.streamlit.app/>

2. How It Works

The app is divided into five pages. You go through them in order, like steps in a process:

Step	Page	What Happens Here
1	Dataset	You upload the student CSV file. The app shows you what the data looks like — number of records, columns, and how many students passed or failed.
2	Preprocess	The app cleans the data — fills in blanks, converts text to numbers, and normalizes values so the models can read it.
3	Train Models	Three models (KNN, SVM, ANN) learn from the data. You can adjust some settings before hitting Train.
4	Evaluate	The app shows you how accurate each model is, so you can see which one did the best job.
5	Predict	You type in a student's details and instantly see what all three models predict for that student.

3. The Dataset

The data used in this system comes from Kaggle, a popular website where people share datasets for data science projects.

File Name	student_performance.csv
Where to get it	https://www.kaggle.com/code/ghadaalaasayedali/student-performance/
Total Students	25,000
Total Columns	16 (15 pieces of information + 1 result)
Result Column	final_grade (a to f) — the app turns this into: Fail, Average, or Pass

What each column means

Here's a plain-language explanation of every column in the dataset:

Column	What It Means
student_id	Just a number to identify each student. The app ignores this.
age	How old the student is (14 to 19 years old).
gender	male, female, or other.
school_type	Whether the student goes to a public or private school.
parent_education	The highest level of education the student's parent finished — from no formal schooling up to a PhD.
study_hours	How many hours per day the student spends studying (0.5 to 8 hours).
attendance_percentage	What percentage of classes the student attended (50% to 100%).
internet_access	Whether the student has internet at home (yes or no).
travel_time	How long it takes the student to get to school (under 15 min up to over 60 min).
extra_activities	Whether the student joins extracurricular activities (yes or no).
study_method	How the student usually studies — notes, textbook, online videos, coaching, group study, or mixed.
math_score	Score in Math (0 to 100).
science_score	Score in Science (0 to 100).
english_score	Score in English (0 to 100).
overall_score	The average of the three subject scores. The app uses this to decide if a student Passes, is Average, or Fails.

final_grade	The letter grade (a to f). The app drops this and uses overall_score instead.
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How the app decides Pass, Average, or Fail

The app looks at the overall_score column and sorts each student into one of three groups:

Label	Score Range	Number of Students
✔ Pass	70 – 100	3,901 (15.6%)
Average	50 – 69	12,472 (49.9%)
✘ Fail	0 – 49	8,627 (34.5%)

4. Cleaning the Data

Before the models can learn anything, the data needs to be cleaned up and prepared. The app does this in four steps:

Step 1 – Fix Missing Values

Sometimes datasets have blanks or missing entries. The app checks for these and lets you choose how to handle them — fill in the blanks with an average number, fill them with the most common value, or just remove those rows entirely. The `student_performance.csv` file has no missing values, so this step is skipped automatically.

Step 2 – Convert Text to Numbers

Prediction models can only work with numbers, not text. So columns like `gender` (male/female/other) or `study_method` (notes/textbook/etc.) get converted into numbers. The app does this automatically.

The result column (`performance`) gets converted too: Fail = 0, Average = 1, Pass = 2.

Step 3 – Normalize the Numbers

Different columns have very different ranges — study hours go from 0 to 8, but scores go from 0 to 100. Normalizing brings everything to a similar scale so the models treat all features fairly.

Step 4 – Split into Training and Testing

The app splits the data into two parts. The models learn from the first part (training set), and then we check how well they do on the second part (test set) — data they've never seen before.

Part	Size (Default)
Training Set (learning)	80% — about 20,000 students
Test Set (checking accuracy)	20% — about 5,000 students

5. The Three Prediction Models

The app trains three different models. Think of each one as a different approach to answering the same question: 'How will this student likely perform?'

K-Nearest Neighbors (KNN)

KNN answers by looking at the most similar students in the dataset and seeing what they got. If your 5 closest matches all passed, it predicts you'll pass too. Simple and easy to understand.

- Main setting: K (how many similar students to look at) — default is 5
- You can also change how it measures 'similarity' between students

Support Vector Machine (SVM)

SVM draws an imaginary boundary that separates Fail, Average, and Pass students as clearly as possible. New students get classified based on which side of the boundary they fall on.

- Main setting: C (how strict the boundary is) — default is 1.0
- You can choose different boundary shapes: RBF, Linear, or Polynomial

Artificial Neural Network (ANN)

ANN is loosely inspired by how the brain works — it passes information through layers of connected nodes that learn patterns in the data. It can pick up on more complex relationships between features.

- Main settings: number of neurons in each of its two hidden layers — defaults are 64 and 32
- Stops training automatically when it stops improving (early stopping)

Quick Comparison

Model	Strength	Things to Keep in Mind
KNN	Easy to understand; works well when similar students behave similarly	Can be slow on very large datasets
SVM	Strong at finding clear boundaries between groups	Harder to tune; slower to train than KNN
ANN	Good at learning complex patterns in big datasets	Needs more data to work well; harder to explain

6. Checking How Accurate the Models Are

After training, the app tests each model on the students it hasn't seen yet. It measures four things:

Metric	What It Means in Plain English
Accuracy	Out of all predictions made, how many were correct? (e.g., 82% correct)
Precision	When the model says a student will Fail, how often is it actually right?
Recall	Out of all the students who actually failed, how many did the model catch?
F1-Score	A single number that balances Precision and Recall — the main score used to pick the best model.

The app also shows a confusion matrix for each model — a grid that shows exactly where the model got things right and where it made mistakes (e.g., predicted Average but the student actually Passed).

At the end of the Evaluate page, the app automatically highlights the best model based on the F1-Score.

7. Making a Prediction

Once the models are trained, go to the Predict page. Fill in the student's details using the sliders and dropdowns, then click the Predict button.

What you fill in

Field	Options or Range
Age	14 to 19
Study Hours Per Day	0 to 8 hours
Attendance	50% to 100%
Math Score	0 to 100
Science Score	0 to 100
English Score	0 to 100
Gender	male, female, other
School Type	public or private
Internet Access	yes or no
Travel Time	< 15 min, 15–30 min, 30–60 min, > 60 min
Extra Activities	yes or no
Parent Education	no formal, high school, diploma, graduate, post graduate, phd
Study Method	notes, textbook, online videos, coaching, group study, mixed

What you get back

All three models give their prediction at the same time. The app then takes a majority vote — if 2 out of 3 models say Pass, the final answer is Pass. You also get a confidence level (e.g., 2/3 or 3/3 models agree).

What the app recommends

Result	Recommendation
✓ Pass	Great job! The student is doing well. Encourage advanced courses, competitions, or peer mentoring.
Average	There's room to grow. Suggest a consistent study schedule, aiming for higher attendance, and joining a study group.
✗ Fail	This student needs help now. Recommend tutoring, a parent-teacher meeting, and an academic support plan.

8. Known Issues and What's Next

Current Limitations

- The dataset mostly reflects general student trends and may not match every local school's grading system.
- The Pass group is small (~15.6%) compared to Fail and Average, so models may be less confident when predicting Pass.
- Trained models are not saved — if you refresh the page, you need to retrain from scratch.

Future Improvements

- Save trained models so you don't have to retrain every session
- Better handle the imbalance between Pass, Average, and Fail student counts
- Add a feature that explains why the model made a specific prediction
- Allow schools to upload their own data with a step-by-step column matching guide
- Track individual students over multiple time periods